Case Study on Google Cloud Platform

Google Cloud Platform enables developers to build, test and deploy applications on Google's highly-scalable and reliable infrastructure. Choose from computing, storage and application services for your web, mobile and backend solutions

Google Compute Engine now generally available

Google Cloud Platform gives developers the flexibility to architect applications with both managed and unmanaged services that run on Google's infrastructure. We've been working to improve the developer experience across our services to meet the standards our own engineers would expect here at Google.

Today, Google Compute Engine is Generally Available (GA), offering virtual machines that are performant, scalable, reliable, and offer industry-leading security features like encryption of data at rest. Compute Engine is available with 24/7 support and a 99.95% monthly SLA for your mission-critical workloads. We are also introducing several new features and lower prices for persistent disks and popular compute instances.

Expanded operating system support
During Preview, Compute Engine supported two of the most popular Linux distributions, Debian and Centos, customized with a Google-built kernel. This gave developers a familiar environment to build on, but some software that required specific kernels or loadable modules (e.g. some file systems) were not supported. Now you can runany out-of-the-box Linux distribution (including SELinux and CoreOS) as well as any kernel or software you like, including Docker, FOG, xfs and aufs. We're also announcing support for SUSE and Red Hat Enterprise Linux (in Limited Preview) and FreeBSD.

Transparent maintenance with live migration and automatic restart
At Google, we have found that regular maintenance of hardware and software infrastructure is critical to operating with a high level of reliability, security and performance. We're introducing transparent maintenance that combines software and data center innovations with live migration technology to perform proactive maintenance while your virtual machines keep running. You now get all the benefits of regular updates and proactive maintenance without the downtime and reboots typically required. Furthermore, in the event of a failure, we automatically restart your VMs and get them back online in minutes. We've already rolled out this feature to our US zones, with others to follow in the coming months.

New 16-core instances
Developers have asked for instances with even greater computational power and memory for applications that range from silicon simulation to running high-scale NoSQL databases. To serve their needs, we're launching three new instance types in Limited Preview with up to 16 cores and 104 gigabytes of RAM. They are available in the familiar standard, high-memory and high-CPU shapes.

Faster, cheaper Persistent Disks
Building highly scalable and reliable applications starts with using the right storage. Our Persistent Disk service offers you strong, consistent performance along with much higher durability than local disks. Today we're lowering the price of Persistent Disk by 60% per Gigabyte and dropping I/O charges so that you get a predictable, low price for your block storage device. I/O available to a volume scales linearly with size, and the largest Persistent Disk volumes have up to 700% higher peak I/O capability. You can read more about the improvements to Persistent Disk in our previous blog post.

10% Lower Prices for Standard Instances
We're also lowering prices on our most popular standard Compute Engine instances by 10% in all regions.

Customers and partners using Compute Engine
In the past few months, customers like , Cooladata, Mendelics, Evite and Wix have built complex systems on Compute Engine and partners
like SaltStack, Wowza, Rightscale, Qubole, Red Hat, SUSE, and Scalrhave joined
our Cloud Platform Partner Program, with new integrations with Compute Engine.

Google's infrastructure

Build on the same infrastructure that allows Google to return billions of search results in milliseconds, serve 6 billion hours of YouTube video per month and provide storage for 425 million Gmail users.

Global network: Google has one of the largest and most advanced computer networks. Google's backbone network has thousands of miles of fiber optic cable, uses advanced software-defined networking and has edge caching services to deliver fast, consistent and scalable performance. In fact, we even laid our own fiber optic cable under the Pacific Ocean.

Redundancy: Multiple points of presence across the globe provides strong redundancy. Your data is automatically mirrored across storage devices in multiple locations.

Cutting-edge computer science: Infrastructure innovation isn't just about hardware. Google has led the industry with innovations in software infrastructure such
as MapReduce, BigTable and Dremel. Today, Google is pushing the next generation of innovation with products such as Spanner and Flume. When you build on Cloud Platform, you get access to Google's technology innovations faster.

Product

Rapidly develop, deploy and iterate your applications without worrying about system administration. Google manages your application, database and storage servers so you don't have to.

Managed services: Let Google worry about database administration, server configuration, sharding and load balancing while you focus on your code. No need to carry a pager or write boilerplate code.

Developer tools and SDKs: Google integrates with familiar development tools like Eclipse and provides API client libraries and a command-line interface, which makes it easy to build the way you want.

Console and administration: See and manage all of your applications from a single console. View the performance of your applications and manage your account and billing with a simple interface.

Services

Virtual machines. Managed platform. Blob storage. Block storage. NoSQL datastore. MySQL database. Big Data analytics. Google Cloud Platform has all the services your application architecture needs.

Compute: Cloud Platform offers both a fully managed platform and flexible virtual machines, allowing you to choose a system that meets your needs. Use App Engine, our Platform-as-a-Service, when you just want to focus on your code and not worry about patching or maintenance. Get access to raw virtual machines with Compute Engine and have the flexibility to build anything you need.

Storage: Google Cloud Platform provides a range of storage services that allow you to maintain easy and quick access to your data. With Cloud SQL andDatastore you get MySQL or schemaless NoSQL databases, while Cloud Storage provides flexible object storage with global edge caching.

Services: Use Google APIs and services to quickly enable a wide range of functionality for your application. You don't need to build these from scratch, just take advantage of easy integration within Cloud Platform.

Some of their customers

Google cloud platform is set of modular cloud-based services that allow you to create anything from simple websites to complex applications.

Snapchat

“App Engine enabled us to focus on developing the application. We wouldn’t have gotten here without the ease of development that App Engine gave us.”

- Bobby Murphy, CTO and co-Founder

Webfilings

“Google App Engine has the breadth and the depth to grow with you. Every 6 months, it gets better. The Google App Engine team knows what you need to make a competitive application.”

- Brett Harper, Director of Product Development

Khan Academy

“If we didn’t have Google App Engine, we’d be spending a lot more time figuring out server setup and working on routers. Our ability to focus on the actual product is the benefit of Google App Engine.”

- Ben Kamens, Lead Developer

Rovio

“Google App Engine allows us to launch games very quickly with teams of one or two developers per game. Because Google manages all the servers, there is little required of us in terms of maintenance.”

- Stefan Hauk, lead server developer for web games

MAG Interactive

"Our rapid growth to 5M Ruzzle players in less than six months required a highly scalable server solution. Google App Engine transformed this huge challenge into a picnic."

- Roger Skagerwall, CEO

Interactions Marketing

"We are always looking for ways to maximize return and minimize investment. BigQuery is the perfect combination. It’s an on-demand, scalable resource."

- Giovanni DeMeo, Vice President of Global Marketing and Analytics